YIHENG SHU (舒意恒)

Telephone: +(1) 614-260-4431; WeChat: shuyh29

E-mail: shu.251@osu.edu ◇ Homepage: https://yihengshu.github.io ◇ Google Scholar

EDUCATION

Ph.D. Student in Computer Science, The Ohio State University — *Columbus, OH, Aug. 2023 - 2028 Expected*
Advised by Prof. Yu Su, OSU NLP Group — GPA: 3.783 / 4.0

Master in Computer Technology, Nanjing University — *Nanjing, China, Sep. 2020 - Jun. 2023*
Advised by Prof. Yuzhong Qu, Websoft Research Group — GPA: 90.96 / 100

Bachelor in Software Engineering, Northeastern University — *Shenyang, China, Sep. 2016 - Jun. 2020*
Advised by Prof. Guibing Guo and Prof. Gang Wu. — GPA: 4.1073 / 5.0 (top 1.7%)

WORK EXPERIENCES

Research Intern, Google Research — *Mountain View, CA, May 2026 - Aug. 2026*
Full-time

Research Intern, Intuit — *Mountain View, CA, May 2025 - Aug. 2025*
Full-time, Research Team

Research Intern, Microsoft Research Asia — *Beijing, China, Feb. 2022 - Feb. 2023*
Full-time, Knowledge Computing Group

RESEARCH INTERESTS

My research explores continual learning and memory for language agents. Specifically, I focus on: 1) developing efficient RAG systems to enhance reasoning and long-term memory capabilities, and 2) integrating these models with complex external environments such as the Web, knowledge graphs, and databases.

PUBLICATIONS

14. [EMNLP'26 Findings] Building Long-Term LLM Memory with Neo-Davidsonian Discourse Units

Sizhe Zhou, Yanzhen Shen, Siru Ouyang, Yuyang Bai, **Yiheng Shu**, Yu Zhang, Jiawei Han

13. [ICLR'26] REMem: Reasoning with Episodic Memory in Language Agents

Yiheng Shu, Saisri Padmaja Jonnalagedda, Xiang Gao, Bernal Jiménez Gutiérrez, Weijian Qi, Kamalika Das, Huan Sun, Yu Su

12. [NeurIPS'25] Mind2Web 2: Evaluating Agentic Search with Agent-as-a-Judge

Boyu Gou*, Zanming Huang*, Yuting Ning*, Yu Gu, Michael Lin, Weijian Qi, Andrei Kopanev, Botao Yu, Bernal Jiménez Gutiérrez, **Yiheng Shu**, Chan Hee Song, Jiaman Wu, Shijie Chen, Hanane Nour Moussa, Tianshu Zhang, Jian Xie, Yifei Li, Tianci Xue, Zeyi Liao, Kai Zhang, Boyuan Zheng, Zhaowei Cai, Viktor Rozgic, Morteza Ziyadi, Huan Sun, Yu Su

11. [ICML'25] From RAG to Memory: Non-Parametric Continual Learning for Large Language Models

Bernal Jiménez Gutiérrez*, **Yiheng Shu***, Weijian Qi, Sizhe Zhou, Yu Su (*: Equal Contribution)

10. [ICLR'25] Navigating the Digital World as Humans Do: Universal Visual Grounding for GUI Agents

Boyu Gou, Ruohan Wang, Boyuan Zheng, Yanan Xie, Cheng Chang, **Yiheng Shu**, Huan Sun, Yu Su

9. [NeurIPS'24] HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models (GitHub stars >4k)

Bernal Jiménez Gutiérrez, **Yiheng Shu**, Yu Gu, Michihiro Yasunaga, Yu Su

8. [EMNLP'24] Middleware for LLMs: Tools Are Instrumental for Language Agents in Complex Environments

Yu Gu, **Yiheng Shu**, Hao Yu, Xiao Liu, Yuxiao Dong, Jie Tang, Jayanth Srinivasa, Hugo Latapie, Yu Su

7. [EACL'24 SRW] Distribution Shifts Are Bottlenecks: Extensive Evaluation for Grounding Language Models to Knowledge Bases

Yiheng Shu, Zhiwei Yu

6. [AAAI'23] Question Decomposition Tree for Answering Complex Questions over Knowledge Bases

Xiang Huang, Sitao Cheng, **Yiheng Shu**, Yuheng Bao, Yuzhong Qu

5. [EMNLP'22] TIARA: Multi-grained Retrieval for Robust Question Answering over Large Knowledge Base

Yiheng Shu, Zhiwei Yu, Yuhan Li, Börje F. Karlsson, Tingting Ma, Yuzhong Qu, Chin-Yew Lin

4. [COLING'22] Logical Form Generation via Multi-task Learning for Complex Question Answering over Knowledge Bases

Xixin Hu, Xuan Wu, **Yiheng Shu**, Yuzhong Qu

3. [ISWC'21] EDG-based Question Decomposition for Complex Question Answering over Knowledge Bases

Xixin Hu, **Yiheng Shu**, Xiang Huang, Yuzhong Qu

2. [TOIS'20] Deep Learning for Sequential Recommendation: Algorithms, Influential Factors, and Evaluations

Hui Fang, Danning Zhang, **Yiheng Shu**, Guibing Guo

1. [ICWE'19 tutorial] Deep Learning-based Sequential Recommender Systems: Concepts, Algorithms, and Evaluations

Hui Fang, Danning Zhang, Guibing Guo, **Yiheng Shu**

SELECTED HONORS & AWARDS

First-class Yingcai Scholarship, Nanjing University	*2022*
First-class Academic Scholarship for Master Students, Nanjing University	*2020 - 2022*
Provincial Excellent Graduates (top 3%), Education Department of Liaoning Province	*2020*
Meritorious Winner in MCM/ICM (international top 7%), COMAP	*2019*
Suzhou Industrial Park Scholarship (top 5%), Northeastern University (China)	*2019*
Pacemaker to Outstanding Student (top 1%), Northeastern University (China)	*2018*
National Scholarship (top 3%), Ministry of Education of the P.R. China	*2018*

SERVICES

Conference reviewer: ARR, ICLR, ICML, NeurIPS, AAAI

Journal reviewer: IEEE Trans. Big Data

Last modified: Oct. 2026